

JADAVPUR UNIVERSITY

188, Raja S.C. Mallick Road, Kolkata – 700032

Department of Computer Science and Engineering, FET.

MASTER OF COMPUTER APPLICATION

1st Year 2nd Semester

DATABASE MANAGEMENT SYSTEM ASSIGNMENT

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Question 1. Create the following tables :

Tablename: EMPLOYEE

Structure :

EMP_CODE char(16)

EMP_NAME char(20)

DEPT_CODE char(16)

DESIG_CODE char(16)

SEX char(1)

ADDRESS char (25)

CITY char (20)

STATE char (20)

PIN char (6)

BASIC Number

JN_DT Date

Primary key is EMP_CODE

Tablename : DESIGNATION

Structure :

DESIG_CODE char(16)

DESIG_DESC char(20)

Primary key is DESIG_CODE

Tablename : DEPARTMENT

Structure :

DEPT_CODE char(16)

DEPT_NAME char(20)

Primary key is DEPT_CODE.

Query :

```
CREATE TABLE IF NOT EXISTS DEPARTMENT (  
    DEPT_CODE CHAR(16) PRIMARY KEY,  
    DEPT_NAME CHAR(20)  
);
```

```
CREATE TABLE IF NOT EXISTS DESIGNATION (  
    DESIG_CODE CHAR(16) PRIMARY KEY,  
    DESIG_DESC CHAR(20)  
);
```

```
CREATE TABLE IF NOT EXISTS EMPLOYEE (  
    EMP_CODE CHAR(16) PRIMARY KEY,  
    EMP_NAME CHAR(20),  
    DEPT_CODE CHAR(16),  
    DESIG_CODE CHAR(16),  
    SEX CHAR(1),  
    ADDRESS CHAR(25),  
    CITY CHAR(20),  
    STATE CHAR(20),  
    PIN CHAR(6),  
    BASIC DECIMAL(10,2),  
    JN_DT DATE,  
  
    FOREIGN KEY (DEPT_CODE) REFERENCES DEPARTMENT(DEPT_CODE),  
    FOREIGN KEY (DESIG_CODE) REFERENCES DESIGNATION(DESIG_CODE)  
);
```

Output :

```
mysql> CREATE TABLE DEPARTMENT (DEPT_CODE CHAR(16) PRIMARY KEY, DEPT_NAME CHAR(20));  
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> CREATE TABLE DESIGNATION (DESIG_CODE CHAR(16) PRIMARY KEY, DESIG_DESC CHAR(20));  
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> CREATE TABLE EMPLOYEE (  
->     EMP_CODE CHAR(16) PRIMARY KEY,  
->     EMP_NAME CHAR(20),  
->     DEPT_CODE CHAR(16),  
->     DESIG_CODE CHAR(16),  
->     SEX CHAR(1),  
->     ADDRESS CHAR(25),  
->     CITY CHAR(20),  
->     STATE CHAR(20),  
->     PIN CHAR(6),  
->     BASIC DECIMAL(10,2),  
->     JN_DT DATE,  
->  
->     FOREIGN KEY (DEPT_CODE) REFERENCES DEPARTMENT(DEPT_CODE),  
->     FOREIGN KEY (DESIG_CODE) REFERENCES DESIGNATION(DESIG_CODE)  
-> );
```

```
Query OK, 0 rows affected (0.05 sec)
```

Question 2 : Display the structure of each table .

Query :

```
DESC department;  
DESC designation;  
DESC employee;
```

Output :

```
mysql> DESC department;
```

Field	Type	Null	Key	Default	Extra
DEPT_CODE	char(16)	NO	PRI	NULL	
DEPT_NAME	char(20)	YES		NULL	

2 rows in set (0.02 sec)

```
mysql> DESC designation;
```

Field	Type	Null	Key	Default	Extra
DESIG_CODE	char(16)	NO	PRI	NULL	
DESIG_DESC	char(20)	YES		NULL	

2 rows in set (0.00 sec)

```
mysql> DESC employee;
```

Field	Type	Null	Key	Default	Extra
EMP_CODE	char(16)	NO	PRI	NULL	
EMP_NAME	char(20)	YES		NULL	
DEPT_CODE	char(16)	YES	MUL	NULL	
DESIG_CODE	char(16)	YES	MUL	NULL	
SEX	char(1)	YES		NULL	
ADDRESS	char(25)	YES		NULL	
CITY	char(20)	YES		NULL	
STATE	char(20)	YES		NULL	
PIN	char(6)	YES		NULL	
BASIC	decimal(10,2)	YES		NULL	
JN_DT	date	YES		NULL	

11 rows in set (0.00 sec)

Question 3: Insert few rows in each table .

[While entering data in EMP table use DESIG_CODE which exists in DESIGNATION table and DEPT_CODE which exists in DEPARTMENT table. In DESIGNATION table, assign code for Manager, Executive, officer, clerk and helper. In DEPARTMENT table, assign code for Personnel, Production, Purchase, Finance, Research departments]

Query :

```
INSERT INTO DESIGNATION VALUES
```

```
('D001','Manager'),  
('D002','Executive'),  
('D003','Officer'),  
('D004','Clerk'),  
('D005','Helper');
```

```
INSERT INTO DEPARTMENT VALUES
```

```
('DP001','Personnel'),  
('DP002','Production'),  
('DP003','Purchase'),  
('DP004','Finance'),  
('DP005','Research');
```

```
INSERT INTO EMPLOYEE VALUES
```

```
('E001','Rahul Das','DP001','D001','M','12 Park  
Street','Kolkata','WB','700001',60000,'2023-01-15'),  
('E002','Anita Roy','DP002','D002','F','45 MG  
Road','Kolkata','WB','700002',45000,'2023-02-10'),  
('E003','Suman Sen','DP003','D003','M','21 Lake  
Road','Siliguri','WB','734001',35000,'2022-07-20'),  
('E004','Priya Dey','DP004','D004','F','10 College  
St','Kolkata','WB','700073',25000,'2024-03-05'),  
('E005','Amit Paul','DP005','D005','M','7 Hill Cart  
Rd','Jalpaiguri','WB','735101',18000,'2024-04-12');
```

Output :

```
mysql> INSERT INTO DEPARTMENT VALUES
```

```
-> ('DP001','Personnel'),  
-> ('DP002','Production'),  
-> ('DP003','Purchase'),  
-> ('DP004','Finance'),  
-> ('DP005','Research');
```

```
Query OK, 5 rows affected (0.01 sec)
```

```
Records: 5 Duplicates: 0 Warnings: 0
```

```
mysql> INSERT INTO DESIGNATION VALUES
```

```
-> ('D001','Manager'),  
-> ('D002','Executive'),  
-> ('D003','Officer'),  
-> ('D004','Clerk'),  
-> ('D005','Helper');
```

```
Query OK, 5 rows affected (0.01 sec)
```

```
Records: 5 Duplicates: 0 Warnings: 0
```

```
mysql> INSERT INTO EMPLOYEE VALUES
```

```
-> ('E001','Rahul Das','DP001','D001','M','12 Park Street','Kolkata','WB','700001',60000,'2023-01-15'),  
-> ('E002','Anita Roy','DP002','D002','F','45 MG Road','Kolkata','WB','700002',45000,'2023-02-10'),  
-> ('E003','Suman Sen','DP003','D003','M','21 Lake Road','Siliguri','WB','734001',35000,'2022-07-20'),  
-> ('E004','Priya Dey','DP004','D004','F','10 College St','Kolkata','WB','700073',25000,'2024-03-05'),  
-> ('E005','Amit Paul','DP005','D005','M','7 Hill Cart Rd','Jalpaiguri','WB','735101',18000,'2024-04-12');
```

```
Query OK, 5 rows affected (0.01 sec)
```

```
Records: 5 Duplicates: 0 Warnings: 0
```


Question 4: In EMP table insert few rows without DEPT_CODE and BASIC.

Query :

```
INSERT INTO EMPLOYEE
(EMP_CODE,EMP_NAME,DESIG_CODE,SEX,ADDRESS,CITY,STATE,PIN,JN_DT) VALUES
('E006','Rita Ghosh','D003','F','22 Station Rd','Malda','WB','732101','2024-05-01'),
('E007','Arjun Roy','D004','M','15 Market Rd','Cooch Behar','WB','736101','2024-06-15');
```

Output :

```
mysql> INSERT INTO EMPLOYEE (EMP_CODE,EMP_NAME,DESIG_CODE,SEX,ADDRESS,CITY,STATE,PIN,JN_DT) VALUES
-> ('E006','Rita Ghosh','D003','F','22 Station Rd','Malda','WB','732101','2024-05-01'),
-> ('E007','Arjun Roy','D004','M','15 Market Rd','Cooch Behar','WB','736101','2024-06-15');
Query OK, 2 rows affected (0.00 sec)
Records: 2 Duplicates: 0 Warnings: 0
```

Question 5: Find the rows with unassigned DEPT_CODE

Query :

```
SELECT * FROM EMPLOYEE WHERE DEPT_CODE IS NULL;
```

Output :

```
mysql> SELECT *
-> FROM EMPLOYEE
-> WHERE DEPT_CODE IS NULL;
```

EMP_CODE	EMP_NAME	DEPT_CODE	DESIG_CODE	SEX	ADDRESS	CITY	STATE	PIN	BASIC	JN_DT
E006	Rita Ghosh	NULL	D003	F	22 Station Rd	Malda	WB	732101	NULL	2024-05-01
E007	Arjun Roy	NULL	D004	M	15 Market Rd	Cooch Behar	WB	736101	NULL	2024-06-15

2 rows in set (0.00 sec)

Question 6 : Find the rows with BASIC equal to zero

Query :

```
SELECT * FROM EMPLOYEE WHERE BASIC = 0;
```

Output :

```
mysql> SELECT *
-> FROM EMPLOYEE
-> WHERE BASIC = 0;
Empty set (0.00 sec)
```

Question 7 : Find the rows with unassigned Basic [note down the difference between the results obtained in Q.6 and Q.7]

Query :

```
SELECT * FROM EMPLOYEE WHERE BASIC IS NULL;
```

Output :

```
mysql> SELECT *
-> FROM EMPLOYEE
-> WHERE BASIC IS NULL;
```

EMP_CODE	EMP_NAME	DEPT_CODE	DESIG_CODE	SEX	ADDRESS	CITY	STATE	PIN	BASIC	JN_DT
E006	Rita Ghosh	NULL	D003	F	22 Station Rd	Malda	WB	732101	NULL	2024-05-01
E007	Arjun Roy	NULL	D004	M	15 Market Rd	Cooch Behar	WB	736101	NULL	2024-06-15

2 rows in set (0.00 sec)

Question 8: Find the average basic of the employees.

Query :

```
SELECT AVG(BASIC) AS Average_Basic FROM EMPLOYEE;
```

Output :

```
mysql> SELECT AVG(BASIC) AS Average_Basic
-> FROM EMPLOYEE;
+-----+
| Average_Basic |
+-----+
| 36600.000000 |
+-----+
1 row in set (0.01 sec)
```

Question 9: Replace the BASIC with 0 for the rows with unassigned Basic.

Query :

```
UPDATE EMPLOYEE SET BASIC = 0 WHERE BASIC IS NULL;
```

Ouput :

```
mysql> UPDATE EMPLOYEE
-> SET BASIC = 0
-> WHERE BASIC IS NULL;
Query OK, 2 rows affected (0.02 sec)
Rows matched: 2 Changed: 2 Warnings: 0
```

Question 10 : Again, find the average Basic. (Note the difference of result obtained in Q.8 & Q.10.)

Query :

```
SELECT AVG(BASIC) AS Average_Basic FROM EMPLOYEE;
```

Output :

```
mysql> SELECT AVG(BASIC) AS Average_Basic
-> FROM EMPLOYEE;
+-----+
| Average_Basic |
+-----+
| 26142.857143 |
+-----+
1 row in set (0.00 sec)
```

Question 11 : Delete the rows with unassigned DEPT_CODE.

Query :

```
DELETE FROM EMPLOYEE WHERE DEPT_CODE IS NULL;
```

Output :

```
mysql> DELETE FROM EMPLOYEE
-> WHERE DEPT_CODE IS NULL;
Query OK, 2 rows affected (0.02 sec)
```

Question 12 : Say, Net pay of an employee= Basic+HRA+DA where HRA is 50% of the Basic & DA is 40% of Basic. Show the employee name & Net pay for all employees.

Query :

```
SELECT EMP_NAME, BASIC + (0.5 * BASIC) + (0.4 * BASIC) AS NET_PAY FROM EMPLOYEE;
```

Output :

```
mysql> SELECT EMP_NAME, BASIC + (0.5 * BASIC) + (0.4 * BASIC) AS NET_PAY FROM EMPLOYEE;
+-----+-----+
| EMP_NAME | NET_PAY |
+-----+-----+
| Rahul Das | 114000.000 |
| Anita Roy | 85500.000 |
| Suman Sen | 66500.000 |
| Priya Dey | 47500.000 |
| Amit Paul | 34200.000 |
+-----+-----+
5 rows in set (0.01 sec)
```

Question 13 : Show the EMP_NAME & BASIC in the ascending order of DEPT_CODE. The employee name must appear in uppercase.

Query :

```
SELECT UPPER(EMP_NAME) AS EMP_NAME, BASIC FROM EMPLOYEE ORDER BY DEPT_CODE ASC;
```

Output :

```
mysql> SELECT UPPER(EMP_NAME) AS EMP_NAME,
-> BASIC FROM EMPLOYEE
-> ORDER BY DEPT_CODE ASC;
+-----+-----+
| EMP_NAME | BASIC |
+-----+-----+
| RAHUL DAS | 60000.00 |
| ANITA ROY | 45000.00 |
| SUMAN SEN | 35000.00 |
| PRIYA DEY | 25000.00 |
| AMIT PAUL | 18000.00 |
+-----+-----+
5 rows in set (0.01 sec)
```

Question 14 : Find the employees who have joined after 1st January 2010.

Query :

```
SELECT * FROM EMPLOYEE WHERE JN_DT > '2010-01-01';
```

Output :

```
mysql> SELECT *
-> FROM EMPLOYEE
-> WHERE JN_DT > '2010-01-01';
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| EMP_CODE | EMP_NAME | DEPT_CODE | DESIG_CODE | SEX | ADDRESS | CITY | STATE | PIN | BASIC | JN_DT |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| E001 | Rahul Das | DP001 | D001 | M | 12 Park Street | Kolkata | WB | 700001 | 60000.00 | 2023-01-15 |
| E002 | Anita Roy | DP002 | D002 | F | 45 MG Road | Kolkata | WB | 700002 | 45000.00 | 2023-02-10 |
| E003 | Suman Sen | DP003 | D003 | M | 21 Lake Road | Siliguri | WB | 734001 | 35000.00 | 2022-07-20 |
| E004 | Priya Dey | DP004 | D004 | F | 10 College St | Kolkata | WB | 700073 | 25000.00 | 2024-03-05 |
| E005 | Amit Paul | DP005 | D005 | M | 7 Hill Cart Rd | Jalpaiguri | WB | 735101 | 18000.00 | 2024-04-12 |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
5 rows in set (0.01 sec)
```


Question 15: Find, how many employees have joined in the month of January?

Query :

```
SELECT COUNT(*) AS JAN_JOINED FROM EMPLOYEE WHERE MONTH(JN_DT) = 1;
```

Output :

```
mysql> SELECT COUNT(*) AS JAN_JOINED
-> FROM EMPLOYEE
-> WHERE MONTH(JN_DT) = 1;

+-----+
| JAN_JOINED |
+-----+
|          1 |
+-----+
1 row in set (0.01 sec)
```

Question 16 : Find the maximum & minimum Basic.

Query :

```
SELECT MAX(BASIC) AS MAX_BASIC, MIN(BASIC) AS MIN_BASIC FROM EMPLOYEE;
```

Output :

```
mysql> SELECT MAX(BASIC) AS MAX_BASIC, MIN(BASIC) AS MIN_BASIC FROM EMPLOYEE;

+-----+-----+
| MAX_BASIC | MIN_BASIC |
+-----+-----+
| 60000.00 | 18000.00 |
+-----+-----+
1 row in set (0.00 sec)
```

Question 17 : Find how many Female employees are there?

Query :

```
SELECT COUNT(*) AS FEMALE_EMPLOYEES FROM EMPLOYEE WHERE SEX = 'F';
```

Output :

```
mysql> SELECT COUNT(*) AS FEMALE_EMPLOYEES FROM EMPLOYEE WHERE SEX = 'F';

+-----+
| FEMALE_EMPLOYEES |
+-----+
|                2 |
+-----+
1 row in set (0.01 sec)
```

Question 18: Replace CITY with existing value converted into uppercase for all rows.

Query :

```
UPDATE EMPLOYEE SET CITY = UPPER(CITY);
```

Output :

```
mysql> UPDATE EMPLOYEE SET CITY = UPPER(CITY);
Query OK, 5 rows affected (0.01 sec)
Rows matched: 5  Changed: 5  Warnings: 0
```

Question 19: Find in how many different cities various employees are residing?

Query :

```
SELECT COUNT(DISTINCT CITY) AS TOTAL_CITIES FROM EMPLOYEE;
```

Output :

```
mysql> SELECT COUNT(DISTINCT CITY) AS TOTAL_CITIES FROM EMPLOYEE;
```

```
+-----+
| TOTAL_CITIES |
+-----+
|           3 |
+-----+
```

```
1 row in set (0.01 sec)
```

Question 20 : Display the employee information in the ascending order of DEPT_CODE and with in a Department, it should be in the descending order of BASIC

Query :

```
SELECT * FROM EMPLOYEE ORDER BY DEPT_CODE ASC, BASIC DESC;
```

Output :

```
mysql> SELECT * FROM EMPLOYEE ORDER BY DEPT_CODE ASC, BASIC DESC;
```

EMP_CODE	EMP_NAME	DEPT_CODE	DESIG_CODE	SEX	ADDRESS	CITY	STATE	PIN	BASIC	JN_DT
E001	Rahul Das	DP001	D001	M	12 Park Street	KOLKATA	WB	700001	60000.00	2023-01-15
E002	Anita Roy	DP002	D002	F	45 MG Road	KOLKATA	WB	700002	45000.00	2023-02-10
E003	Suman Sen	DP003	D003	M	21 Lake Road	SILIGURI	WB	734001	35000.00	2022-07-20
E004	Priya Dey	DP004	D004	F	10 College St	KOLKATA	WB	700073	25000.00	2024-03-05
E005	Amit Paul	DP005	D005	M	7 Hill Cart Rd	JALPAIGURI	WB	735101	18000.00	2024-04-12

```
5 rows in set (0.00 sec)
```